

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: medium

Benchmark: MILU | Context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

For the MCQ-label correctness narrowly, MILU has reasonable provenance: answers are inherited from online exam portals where 'subject experts ensure the accuracy of the answers' **[Q26]**, with AI4Bharat volunteer manual audits **[Q87]** and researcher review of subject-tag clusters **[Q41]**. However, the paper reports no annotator demographics, no inter-annotator agreement statistics, and no formal QA protocol beyond manual sampling. Critically, the deployment's core output (Hindi rationale) has no ground-truth labels in MILU at all. Dataset analysis surfaces additional concerns: cross-language item duplication **[D14, D15, D29]** and malformed items with truncated stems **[DATASET-D34, D35]** suggest QA imperfections at the item level. ~55% of subject labels were assigned via machine translation + GPT-4O-MINI **[Q37, Q38]**, an automated process that may not preserve Hindi-specific terminology fidelity.

ID	Evidence
Q26	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q37	"Upon examination, we found that approximately 45% of questions were accurately labeled with a topic name, while the remaining questions lacked this information." (p.4)
Q38	"To address this issue, we first translate the untagged questions into English using INDICTRANS2 (Gala et al., 2023) and then prompt GPT-4O-MINI model to assign an appropriate topic name to the question." (p.4)
Q41	"We manually review these clusters and assign appropriate subject labels." (p.4)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
D14	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? ... Sharp rise in value of imports of oil & petroleum products — English original; same question appears in Hindi (D15) and Bengali — confirms translation pipeline
D15	1991 में वित्तीय संकट को दूर करने के लिए विदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? ... तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of D14 — fluent Hindi, but `is_translated: True`, confirms translation pipeline
D29	ഭരണീയ അടിയന്തരാവസ്ഥ പര്യാപ്തമാക്കുന്നത് 'സാധു കലാപം' എന്ന് പദം ഭരണഘടനയിൽ ചേർത്തത് എപ്പോൾ? ... 44-ാമത് ഭരണഘടനാ ഭേദഗതി നിയമം വഴി — Same Constitutional Amendment question appears in Hindi (D1), Tamil, and Malayalam — cross-language duplication
D34	The owner of the textile shop brought a ... Calculator — Incomplete stem — "brought a" with no further context; question appears malformed
D35	ദിശകൾ: മാർബിൾ ഉപയോഗിക്കാവുന്നത് ... ശിൽപ്പം — Incomplete stem ("Directions: marble can be used") — same truncated structure appears across languages

Strengths

- Answer keys derive from established Indian competitive-exam ecosystems where subject experts validate correctness at source **[Q26]**.
- Multi-stage manual review applied to subject-tag clusters **[Q41]** and a manual audit pass **[Q87]**.
- Constitutional, historical, and GK answer labels in the observed Hindi sample appear factually correct **[DATASET-D1, D2, D22]**.

ID	Evidence
----	----------

Q26	“These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers.” (p.3)
Q41	“We manually review these clusters and assign appropriate subject labels.” (p.4)
Q87	“We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits.” (p.9)
D1	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? ... 44वें संविधान संशोधन अधिनियम द्वारा — Polity/Constitution question in Hindi, about 44th Constitutional Amendment — UPSC core topic
D2	मुगल ग्रंथ 'मासिर-ए-आलमगीरी' के रचयिता कौन हैं? ... साकी मुस्तईद खान — Mughal-period history question in Hindi — directly relevant to North India medieval history
D22	What was the first capital of the Bahamani Kingdom? ... Gulbarga — India-specific historical knowledge (Deccan Sultanate) — same question appears across all languages

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: MCQ answer labels reflect Indian exam-portal subject-expert consensus [Q26], which is reasonable for central-exam content. However, regional stakeholder perspectives on rationale quality have no representation since rationale labels do not exist.

OC-2: INSUFFICIENT DOCUMENTATION on annotator-population disagreement risk for MCQ labels; annotator demographics are not reported beyond 'AI4Bharat volunteers' [Q87].

OC-3: Paper does not provide a Datasheet or Data Statement with annotator demographics — gap for assessing convergent validity.

OC-4: Re-annotation by Hindi-medium subject experts would be needed to (a) validate translation quality on the ~25%+ translated Hindi subset, (b) flag items where the answer is region-sensitive, (c) generate ground-truth Hindi rationales.

OC-5: INSUFFICIENT DOCUMENTATION on aggregation methods; the paper describes manual cluster review [Q41] but not how disagreements were resolved.

OC-6: Label issues: (a) no IAA statistics; (b) ~55% of topic labels assigned by automated MT+GPT pipeline [Q37, Q38]; (c) cross-language duplicates introduce identical labels across translated copies [D14, D15]; (d) malformed items have labels whose validity is questionable [DATASET-D34, D35]; (e) rationale ground-truth wholly absent.

ID	Evidence
Q26	“These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers.” (p.3)
Q37	“Upon examination, we found that approximately 45% of questions were accurately labeled with a topic name, while the remaining questions lacked this information.” (p.4)
Q38	“To address this issue, we first translate the untagged questions into English using INDICTRANS2 (Gala et al., 2023) and then prompt GPT-4O-MINI model to assign an appropriate topic name to the question.” (p.4)
Q41	“We manually review these clusters and assign appropriate subject labels.” (p.4)

Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
D14	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? ... Sharp rise in value of imports of oil & petroleum products — English original; same question appears in Hindi (D15) and Bengali — confirms translation pipeline
D15	1991 में वित्तीय संकट को दूरगिर करने वाले वदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? ... तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of D14 — fluent Hindi, but `is_translated: True`, confirms translation pipeline
D34	The owner of the textile shop brought a ... Calculator — Incomplete stem — "brought a" with no further context; question appears malformed
D35	ദിശകൾ: മാർബിൾ ഉപയോഗിക്കാവുന്നത് ... ശിരപ്പം — Incomplete stem ("Directions: marble can be used") — same truncated structure appears across languages

Information Gaps

- Annotator demographics, IAA statistics, and disagreement-resolution protocols are not reported.
- Translation-quality verification for the Hindi partition is not documented.
- No Hindi rationale ground truth exists in the benchmark.

Requires Expert Verification

- A Hindi-medium subject-matter expert should audit a representative Hindi sample for label correctness, especially for translated items and items with stems that may have been corrupted by RC-passage stripping.

Remediation

Gap 1: No annotator demographics, IAA statistics, or formal QA protocol reported; ~55% of topic labels assigned by automated GPT-4O-MINI [Q37, Q38]; malformed items observed [DATASET-D34, D35].

Recommendation: Conduct a Hindi-medium expert audit on a stratified sample of the Hindi test split: verify answer correctness, translation-register quality, item completeness, and subject-tag accuracy. Report IAA. Filter or repair malformed items prior to deployment-grade evaluation.

ID	Evidence
Q37	"Upon examination, we found that approximately 45% of questions were accurately labeled with a topic name, while the remaining questions lacked this information." (p.4)
Q38	"To address this issue, we first translate the untagged questions into English using INDICTRANS2 (Gala et al., 2023) and then prompt GPT-4O-MINI model to assign an appropriate topic name to the question." (p.4)
D34	The owner of the textile shop brought a ... Calculator — Incomplete stem — "brought a" with no further context; question appears malformed
D35	ദിശകൾ: മാർബിൾ ഉപയോഗിക്കാവുന്നത് ... ശിരപ്പം — Incomplete stem ("Directions: marble can be used") — same truncated structure appears across languages